

Affinity landscape of the intra-tumoral B-cells in lung adenocarcinoma

THOMAS DUPIC

Which antigens are targeted by intra-tumoral B cells?



Affinity landscape of the intra-tumoral B-cells in lung adenocarcinoma

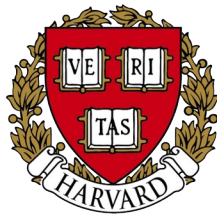
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Post-Doc
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T. Mora, Paris
*Statistics of the
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6 pub., 2 1st aut.

LPENS
LABORATOIRE DE PHYSIQUE
DE L'ÉCOLE NORMALE SUPÉRIEURE



2018

2020

2025



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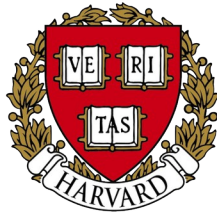
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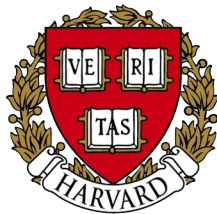
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Team Leader at CIML, Marseille
*Affinities within the immune
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Hybrid team, experimental and
computational

Ciml
IMMUNOLOGY

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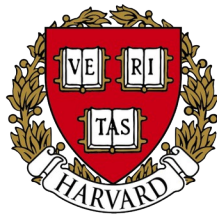
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CSS 2: Cancer &
Genetic diseases

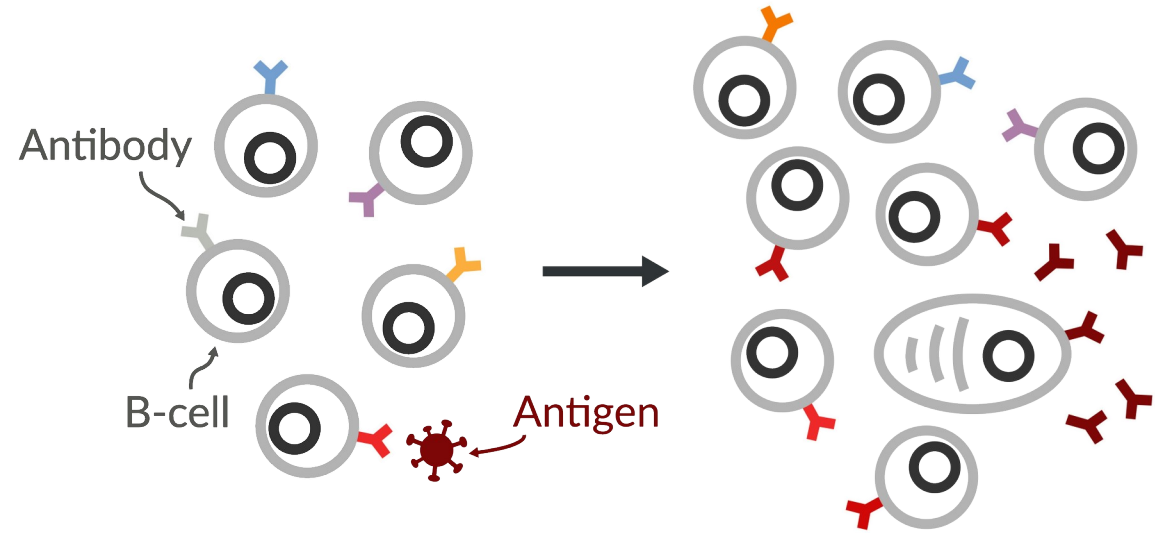
Inserm

Immune repertoires

Immune repertoire : set of all antibodies / T-cell receptors in one individual ($\approx 10^{13}$).

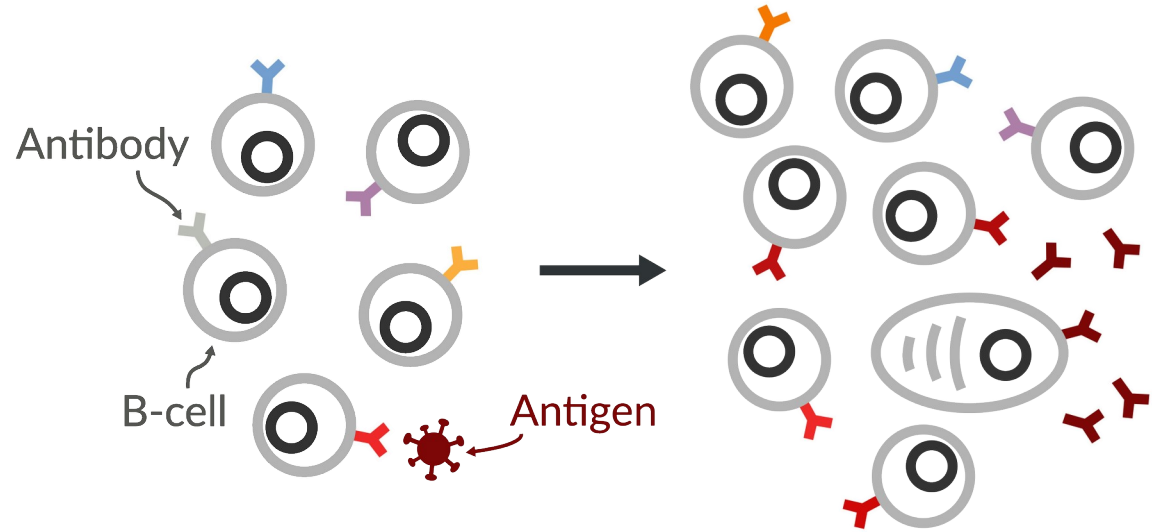
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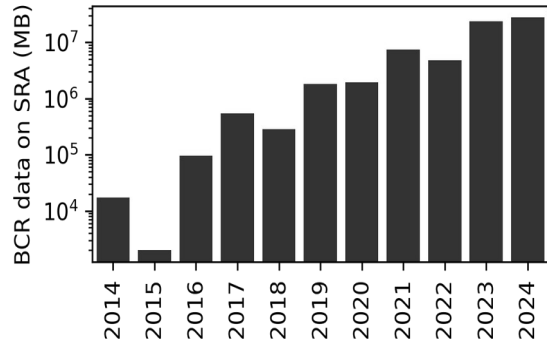
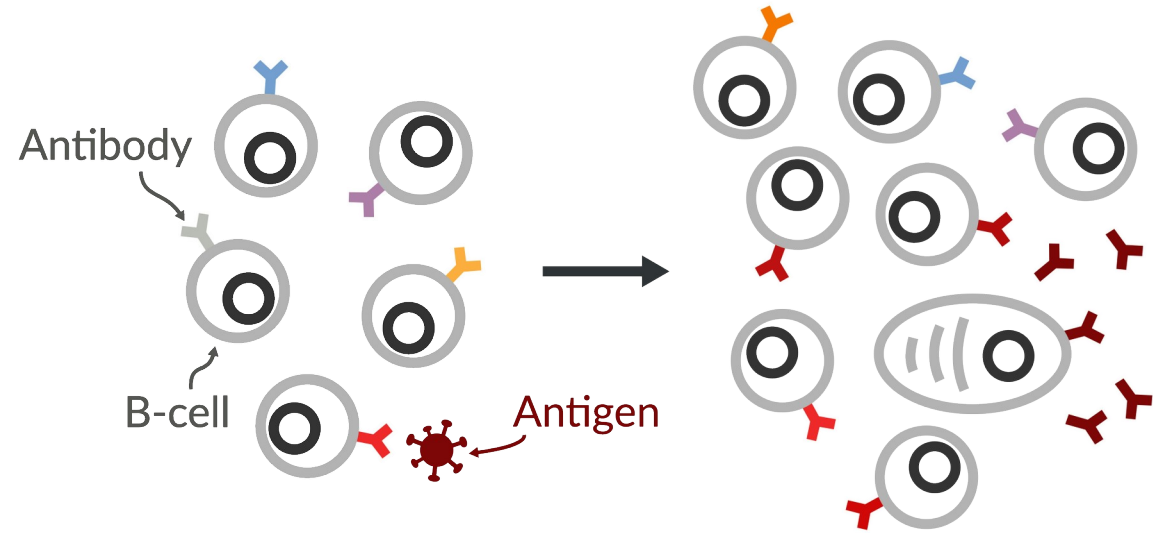
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The repertoire as a prognosis tool ?

Immune repertoires

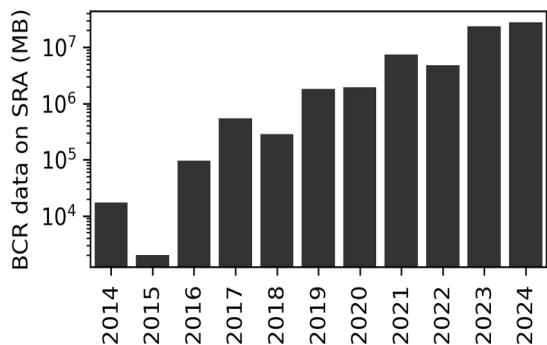
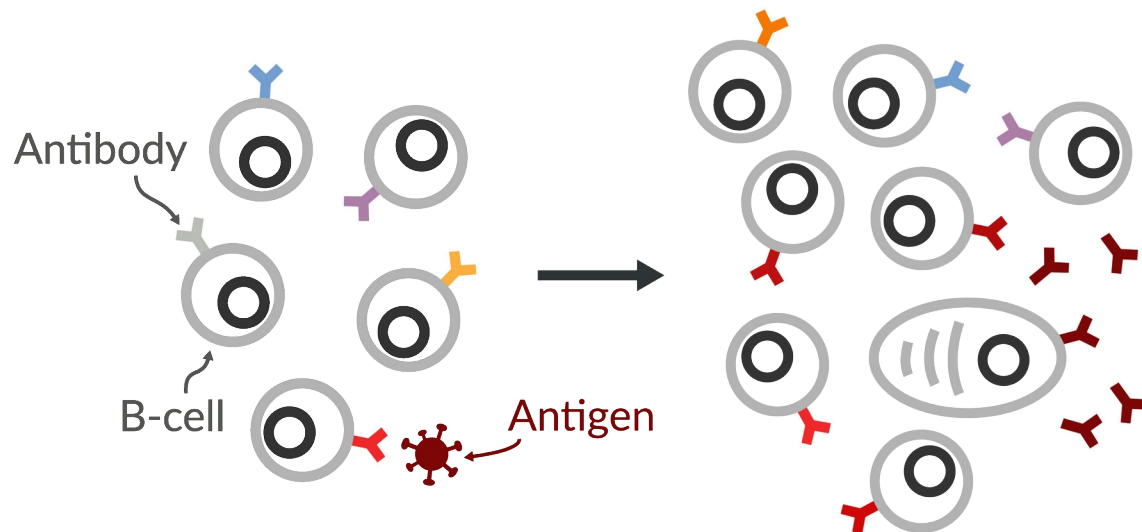
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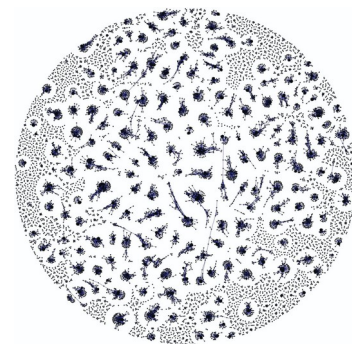
The repertoire as a prognosis tool ?
Easy to access...

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The repertoire as a prognosis tool ?
Easy to access... but hard to interpret.



Immune repertoires

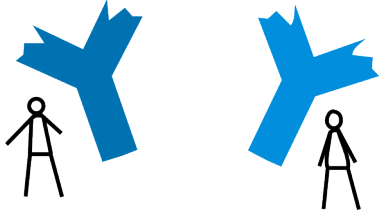
Statistical models, **generation**¹ & **selection**² of the receptors. **Dupic et al., PLOS Comp. Bio (2019)**

$$P(\text{Y}) = \dots$$

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Public repertoire: do shared receptors imply a shared response?

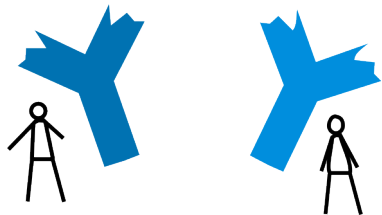
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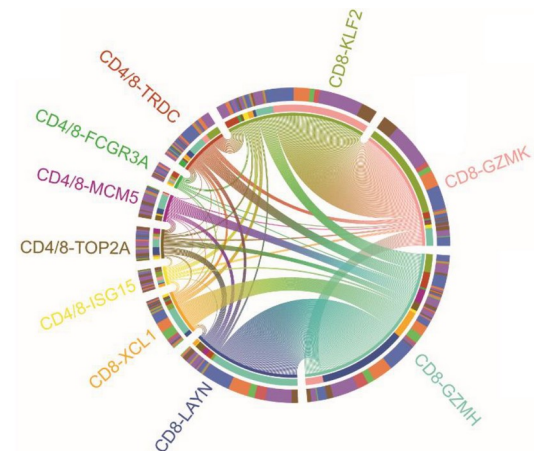
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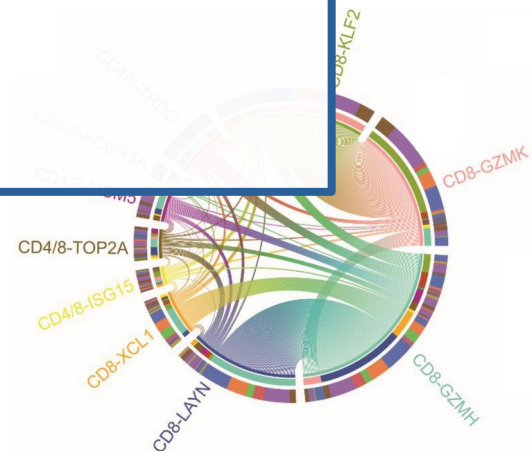
The receptor is used as a **barcode** to track a clone, to estimate diversity.

shared Response?

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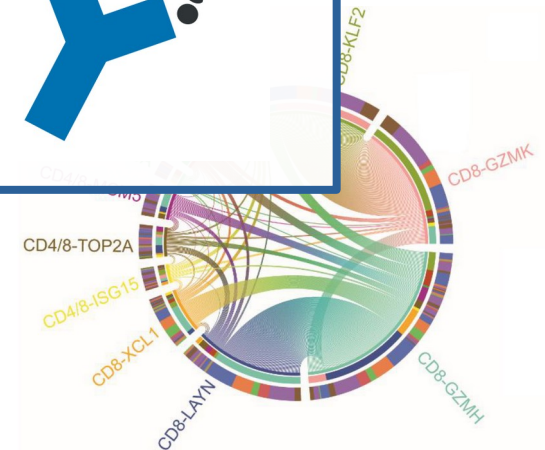
Current methods don't integrate the **functional properties** of the receptor.



Dupic et al.,
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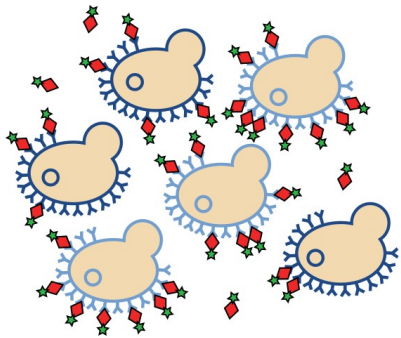
Beyond barcodes : tracking interactions between proteins in high-throughput

Tite-Seq: high-throughput and quantitative method can measure up to 100'000 proteins against one ligand.



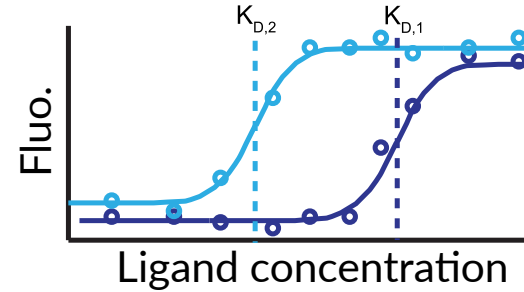
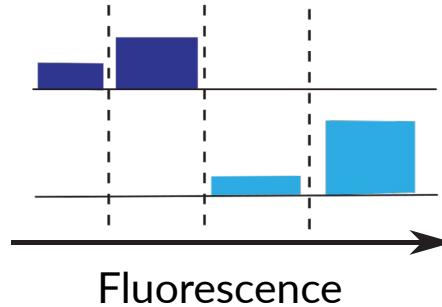
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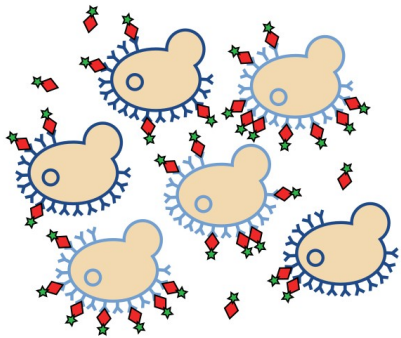
Expression on the yeast with fluorescent antigen

Cell sorting, sequencing and statistics



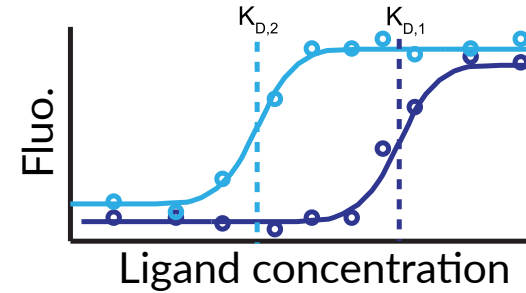
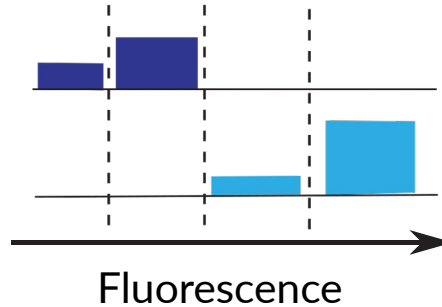
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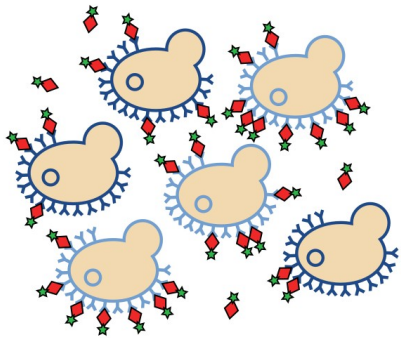
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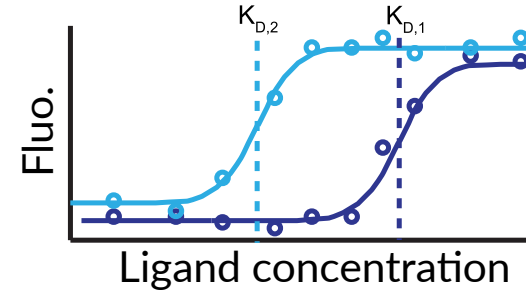
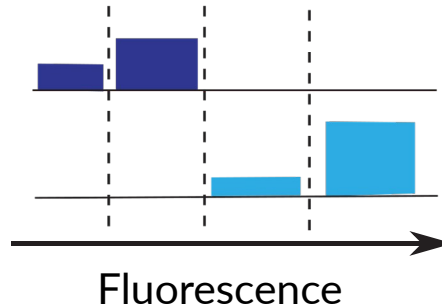
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Antibodies

Phillips, ..., Dupic et al., eLife (2021)

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Dupic, et al., Genetics (2026, in revision)



Measuring affinity at repertoire scale



Measuring affinity at repertoire scale

Long-term B-cell
response to SARS-
CoV-2 vaccination

Moulana, ..., Dupic*, Desai* (in preparation)



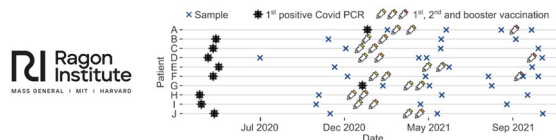
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Repertoire Sequencing →



Common BCRs →

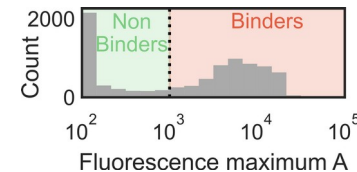


Affinity

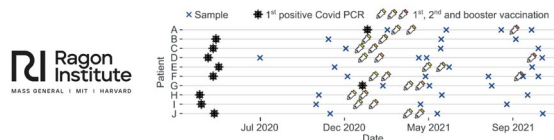
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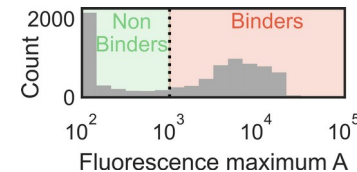
Affinity

1M BCRs tested $\Rightarrow$ 5000 binders

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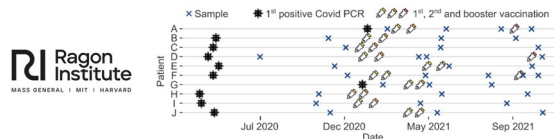


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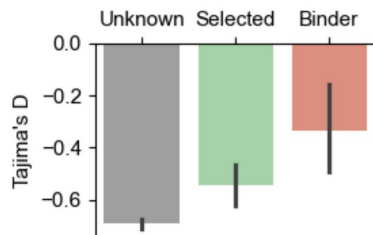
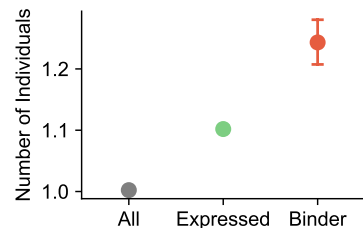
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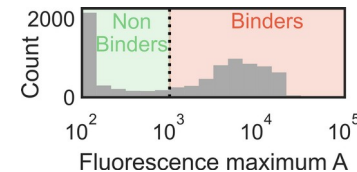


The repertoire statistics of **binders**
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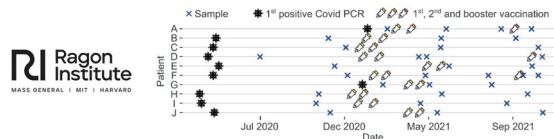
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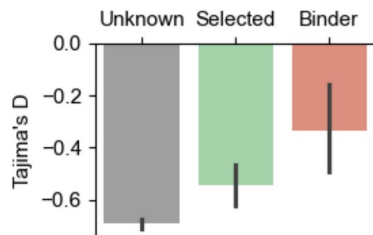
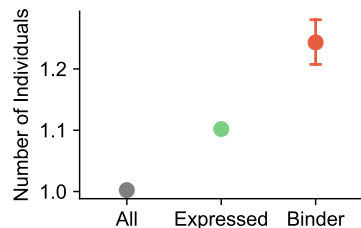
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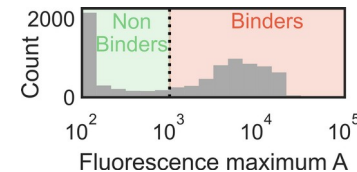
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Improving binding models
with transfer learning

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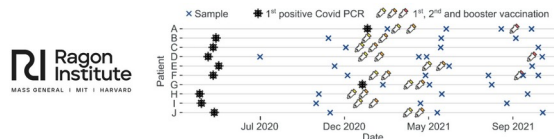


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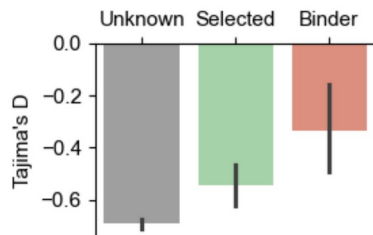
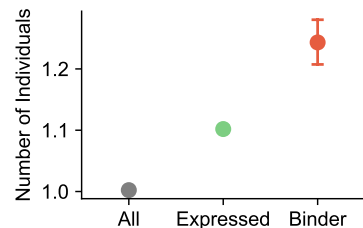
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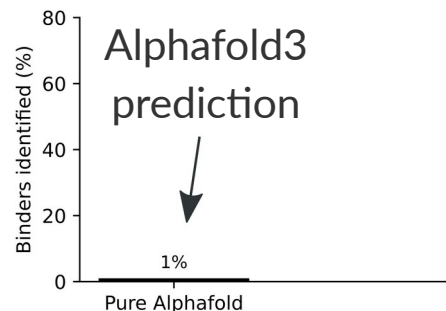
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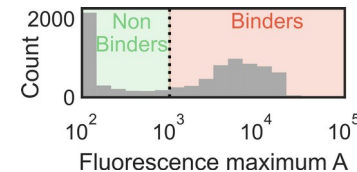
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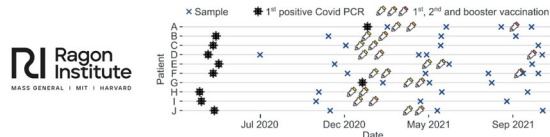


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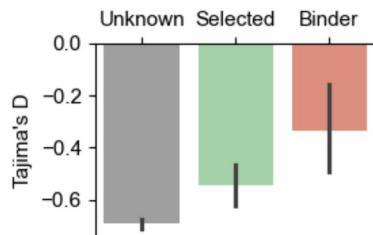
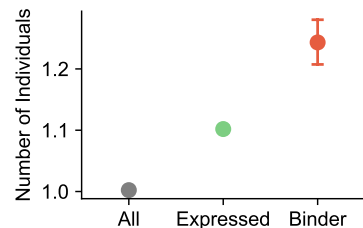
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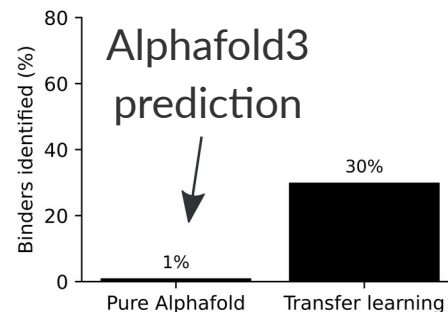
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AlphaFold3 embedding
 $\times$
Our measurements

Measuring affinity at repertoire scale

Tite-Seq × repertoire :



Sequenced
repertoires

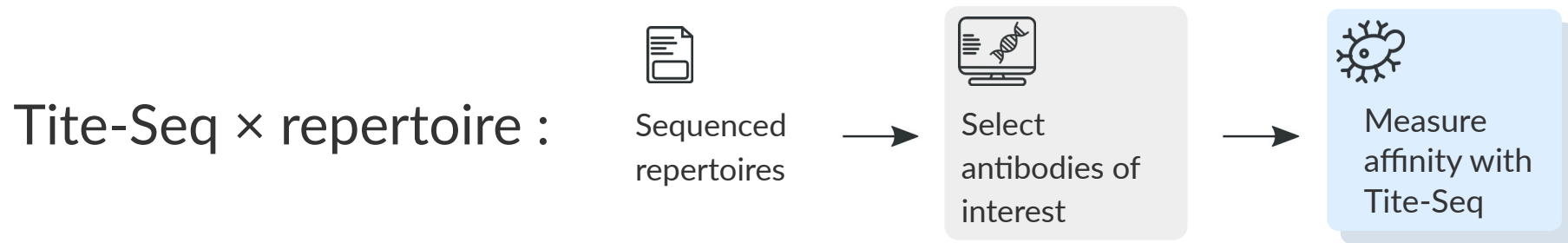


Select
antibodies of
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Measure
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Measuring affinity at repertoire scale



Tite-Seq is precise, mature and cheap

Measuring affinity at repertoire scale

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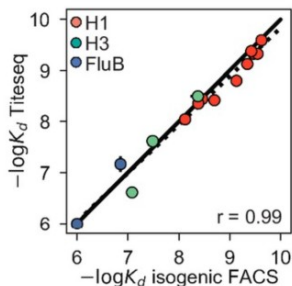


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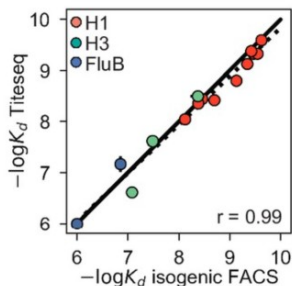


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Starr et al., Nature, 2021
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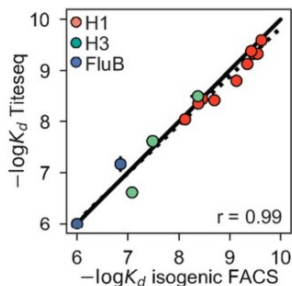


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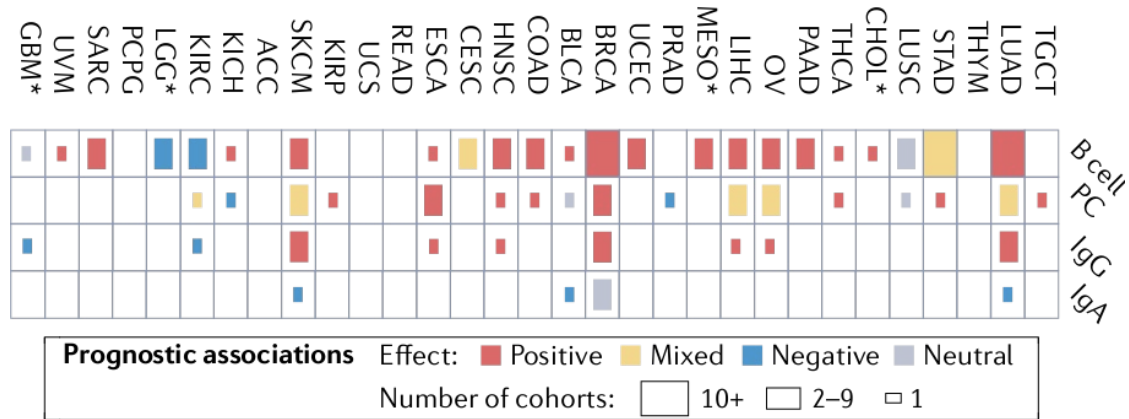


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Tumor Infiltrated B-Lymphocytes (TIL-B)

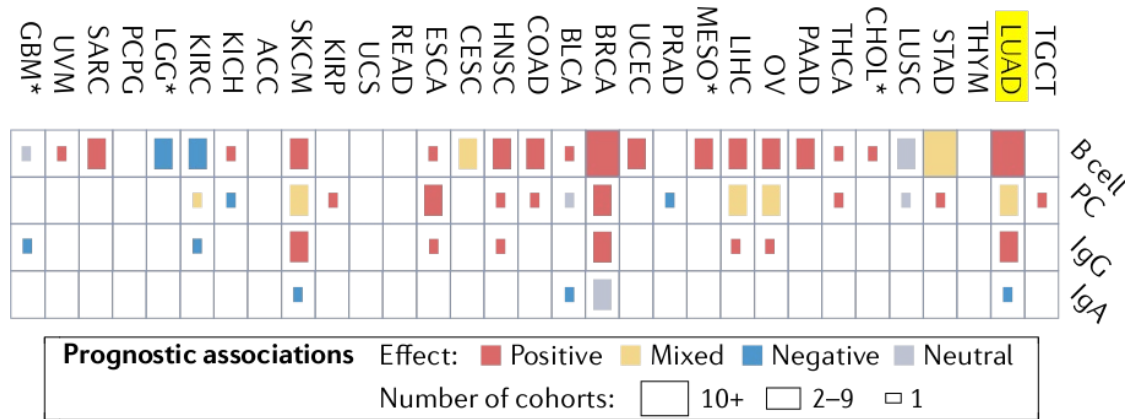
TIL-Bs play a complex role in tumors but generally improve prognosis and response to immunotherapy.



Adapted from *Laumont et al., Nat. Rev. Cancer (2022)*

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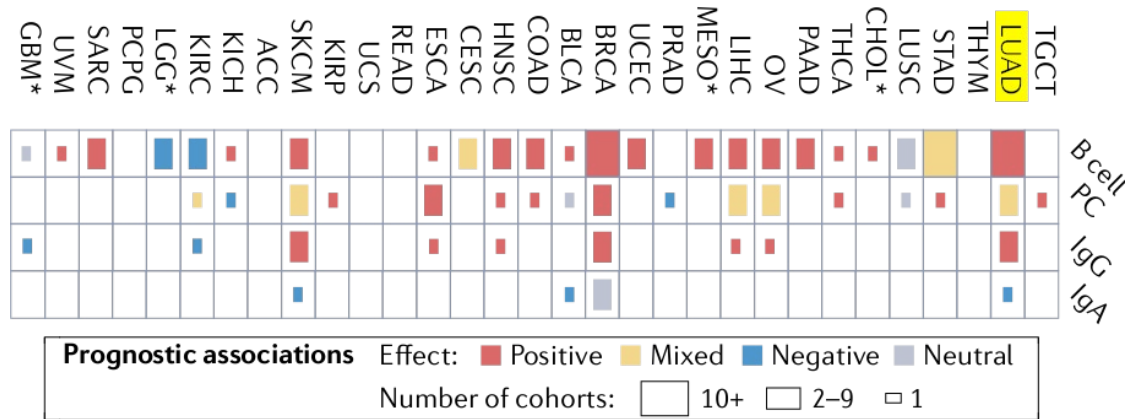
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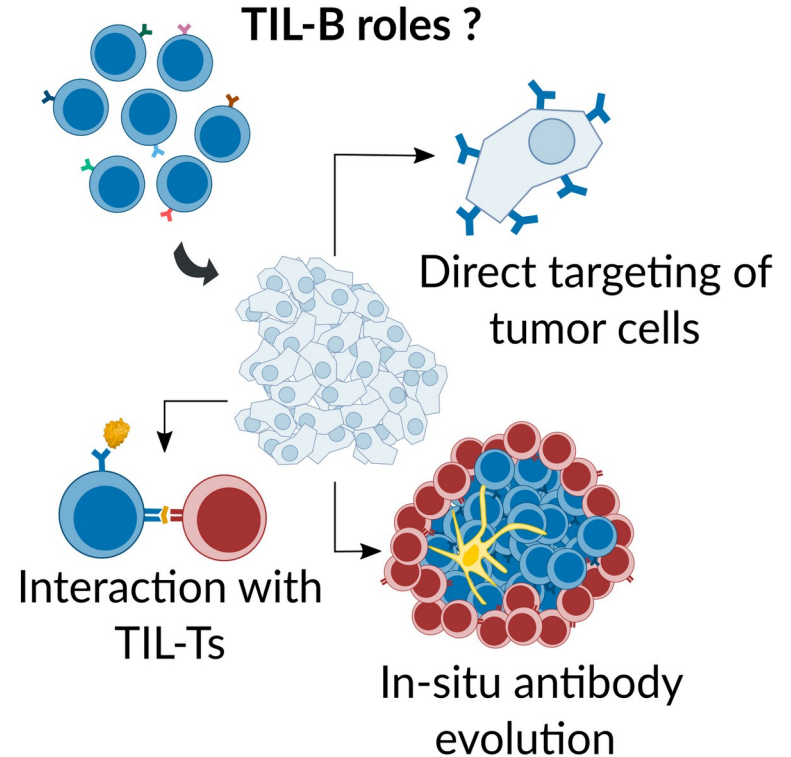
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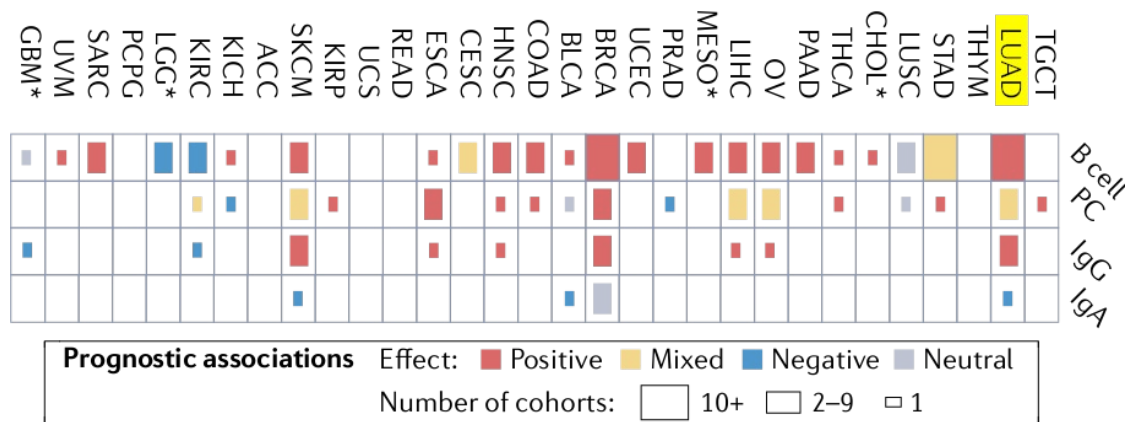


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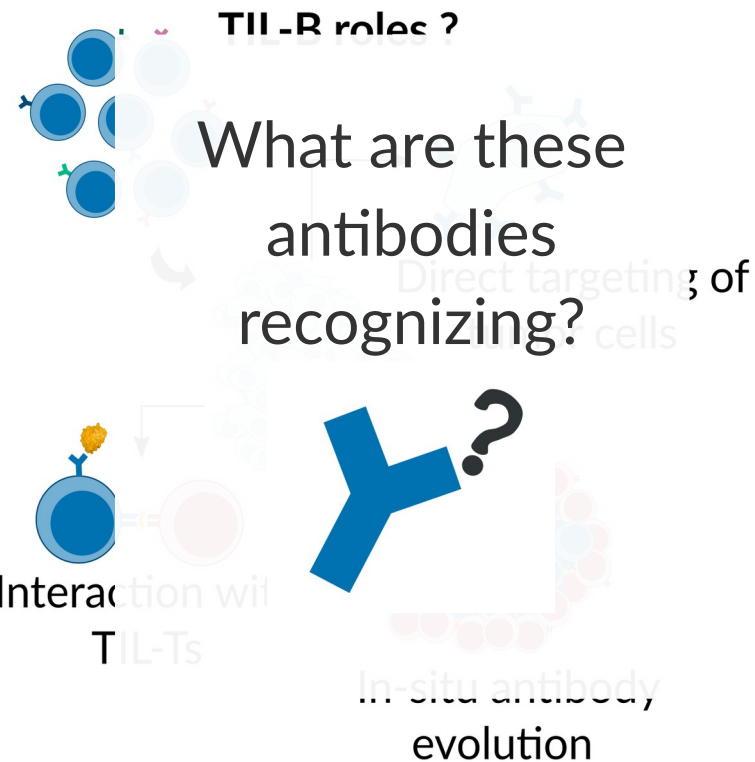


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The public TIL-B repertoire

Only **70 pairs** antigen /
TIL-antibody identified.

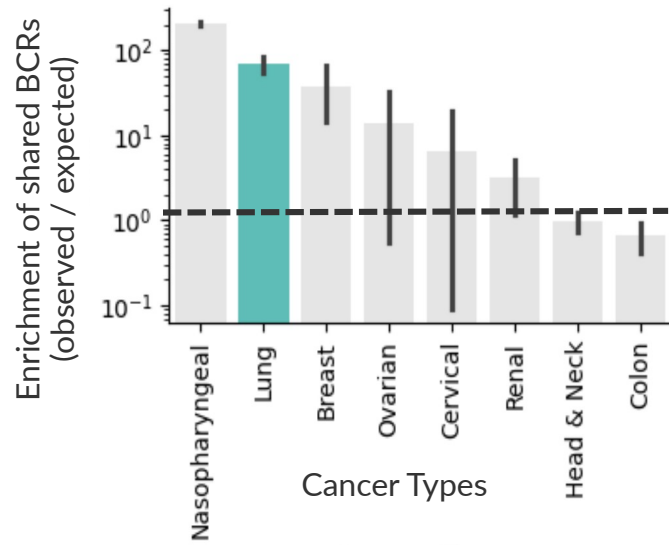
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The intra-tumoral repertoire is
public $\Rightarrow$ directed response !



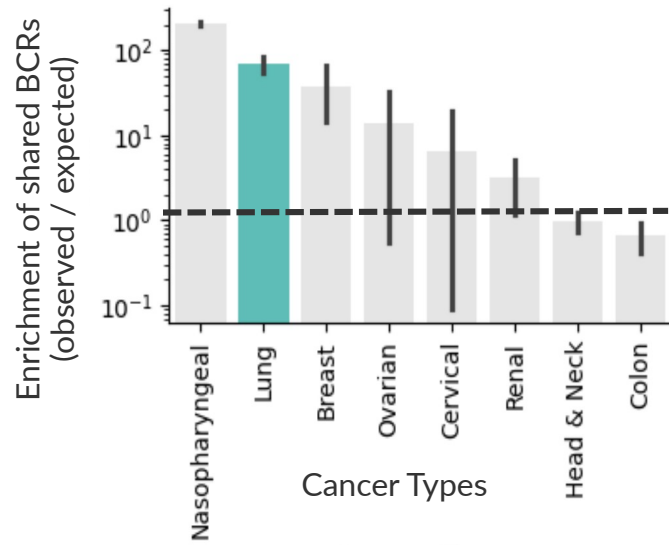
Data from: Aran et al, *Front. In Immunol.*, 2021, McDaniel et al., *Cancer Immunol Immunother*, 2018, Liu et al., *Nat. Commun.*, 2024, Shu et al., *BiorXiv*, 2023, Ma et al., *Science*, 2024, Ruffin et al., *Nat. Commun.*, 2024

The public TIL-B repertoire

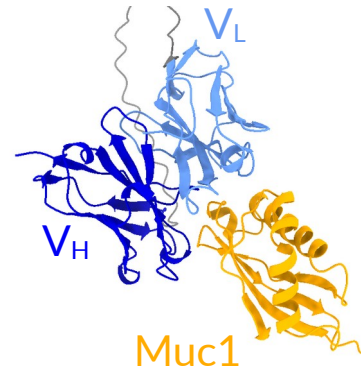
Only **70 pairs** antigen /
TIL-antibody identified.

Laumont et al., Nat. Rev. Cancer (2022)

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Preliminary Data

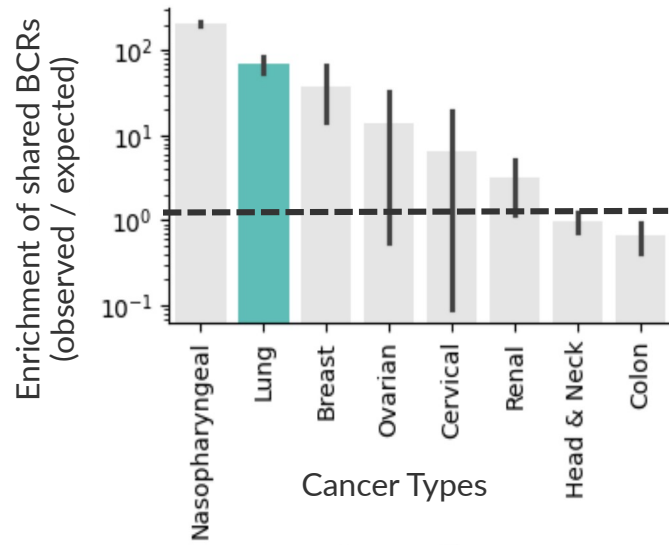
Lung cancer:
500 BCRs tested
in-silico against 5
antigens

The public TIL-B repertoire

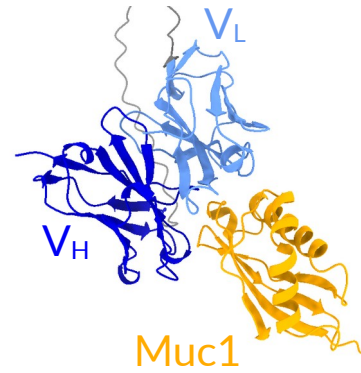
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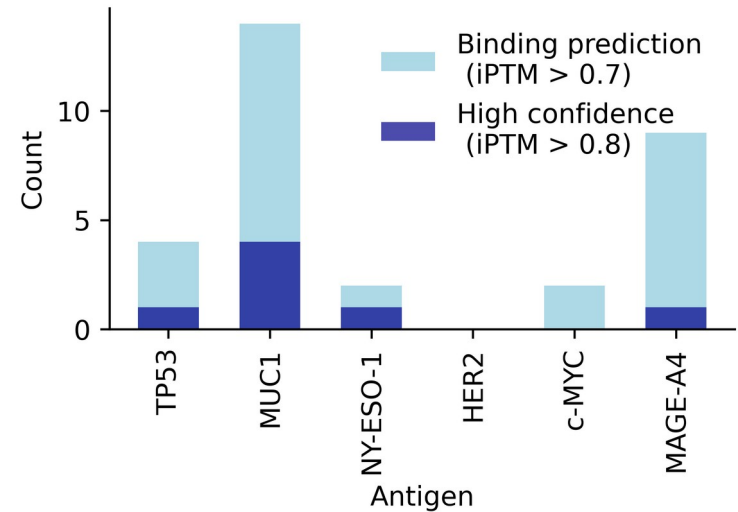


Data from: Aran et al, Front. In Immunol., 2021, McDaniel et al., Cancer Immunol Immunother, 2018, Liu et al., Nat. Commun., 2024, Shu et al., BiorXiv, 2023, Ma et al., Science, 2024, Ruffin et al., Nat. Commun., 2024



Preliminary Data

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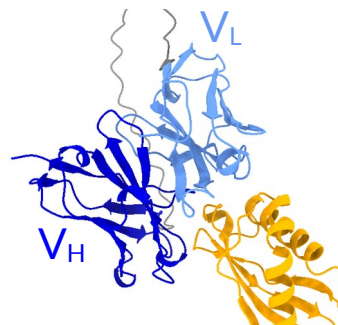
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Preliminary Data

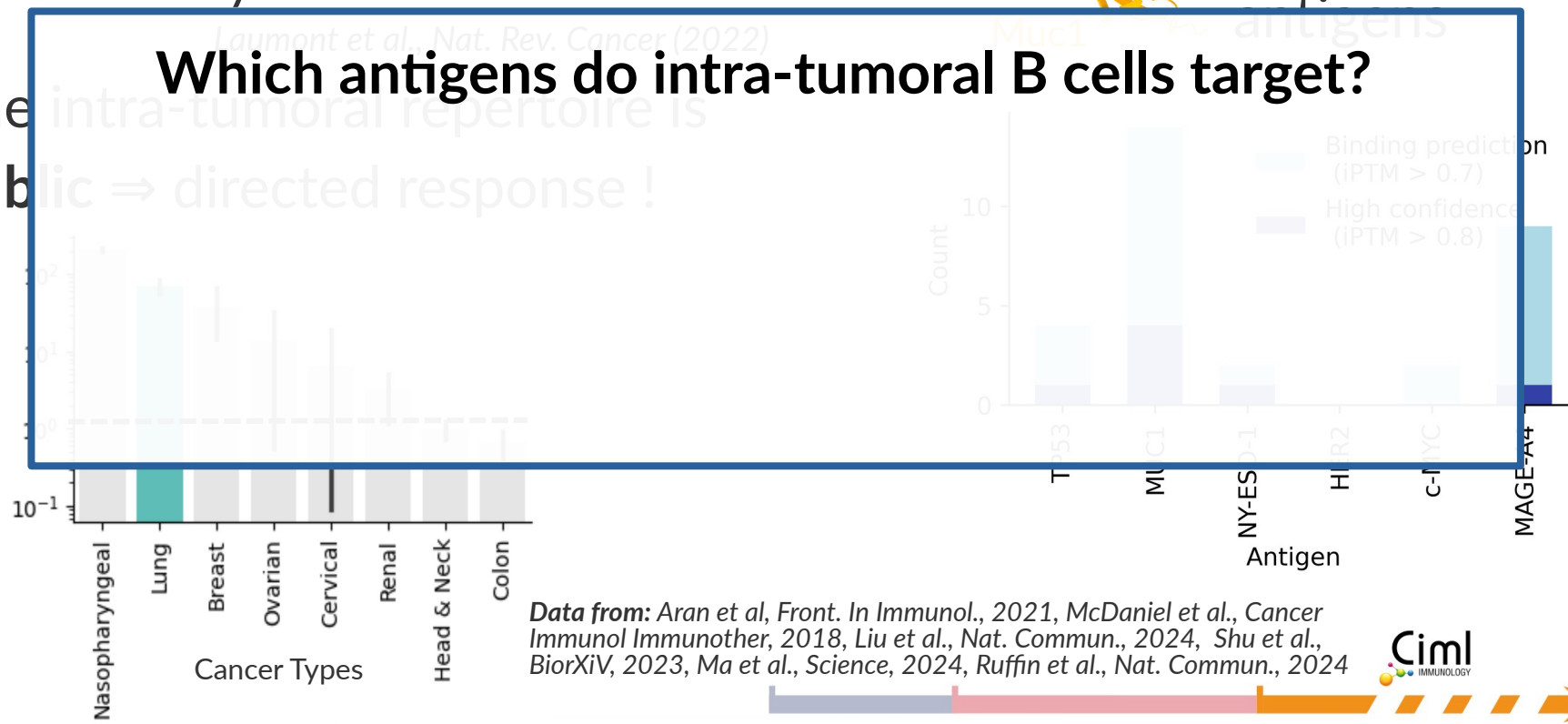
Lung cancer:

500 BCRs tested
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Which antigens do intra-tumoral B cells target?

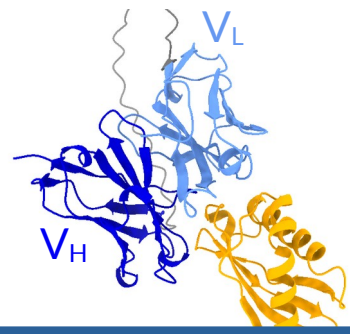
Enrichment of shared BCRs
(observed / expected)



Data from: Aran et al, Front. In Immunol., 2021, McDaniel et al., Cancer Immunol Immunother, 2018, Liu et al., Nat. Commun., 2024, Shu et al., BiorXiv, 2023, Ma et al., Science, 2024, Ruffin et al., Nat. Commun., 2024

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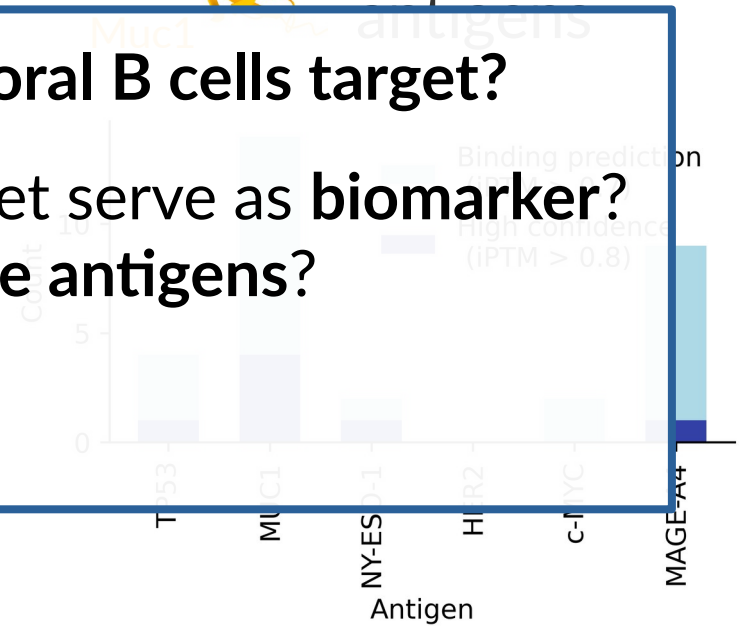
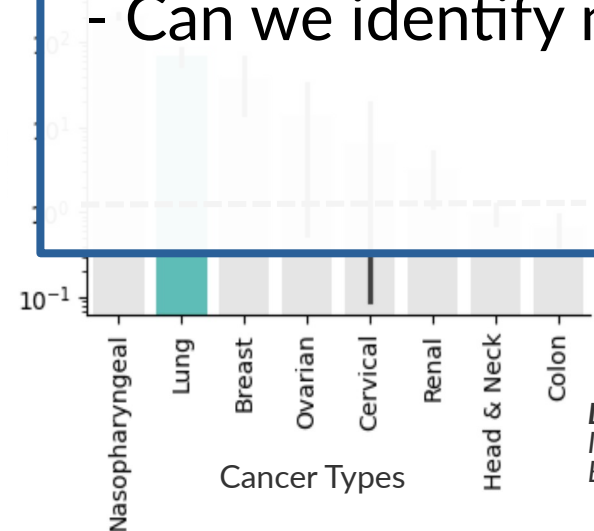


Lung cancer:
500 BCRs tested
in-silico against 5

Which antigens do intra-tumoral B cells target?

- Can the antigens the BCRs target serve as **biomarker**?
- Can we identify novel **targetable** antigens?

Enrichment of shared BCRs
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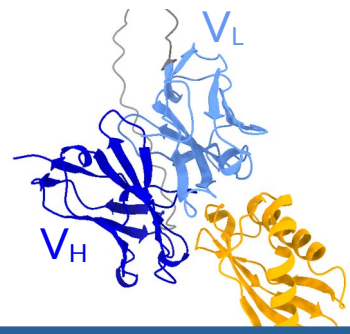


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The public TIL-B repertoire

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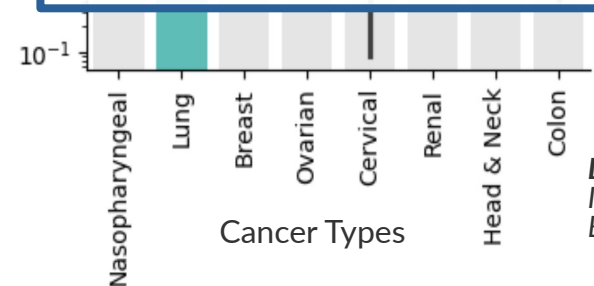
Lung cancer:

500 BCRs tested in-silico against 5

Which antigens do intra-tumoral B cells target?

- Can the antigens the BCRs target serve as **biomarker**?
- Can we identify novel **targetable** antigens?
- Do plasma cells have different specificity?
- Does specificity change during affinity maturation?

Enrichment of shared BCRs (observed / expected)



Cancer Types

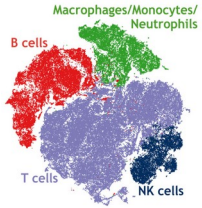
Data from: Aran et al, Front. In Immunol., 2021, McDaniel et al., Cancer Immunol Immunother, 2018, Liu et al., Nat. Commun., 2024, Shu et al., BiorXiv, 2023, Ma et al., Science, 2024, Ruffin et al., Nat. Commun., 2024



Which antigens are targeted by TIL-Bs



Non-small cell lung
cancer dataset(s)



Public data

Hao (2022)

Prazanowska (2023)

...

RHU
LUCA-PI

900 prospective patients
(pulmonary nodules,
PBMC)



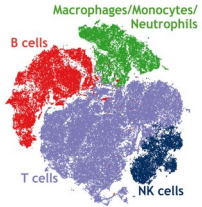
216 patients
(tumor & PBMC
samples)



Which antigens are targeted by TIL-Bs



Non-small cell lung cancer dataset(s)



Public data

Hao (2022)

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...

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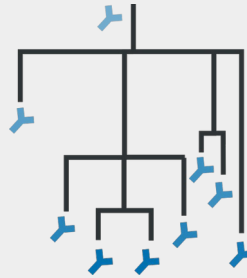
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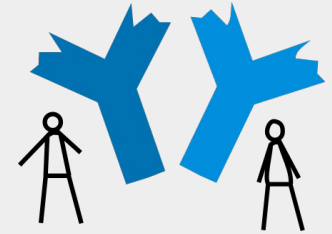


Identify TIL-B of interest *in-silico*
(WP2)



Clonality, isotypes &
transcriptomics

Plasma / Memory B-cells.
Clonal lineages under
selection



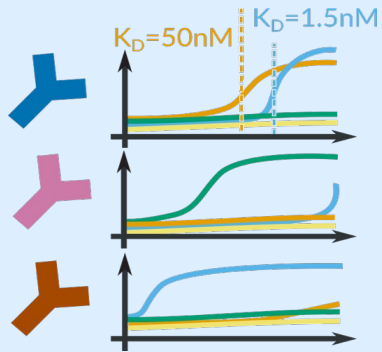
Comparisons
between patients

Identify similar receptors
between repertoires

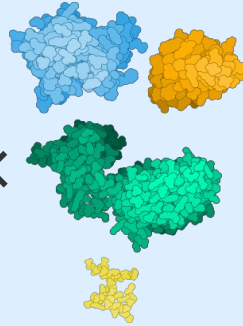
Which antigens are targeted by TIL-Bs



Measure the specificity of the repertoire *ex-vivo* (WP1/WP3)



×



Neoantigens
(EGFR L858R,
EML4-ALK...)

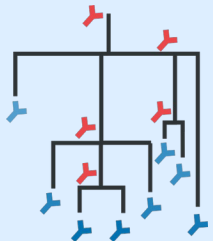
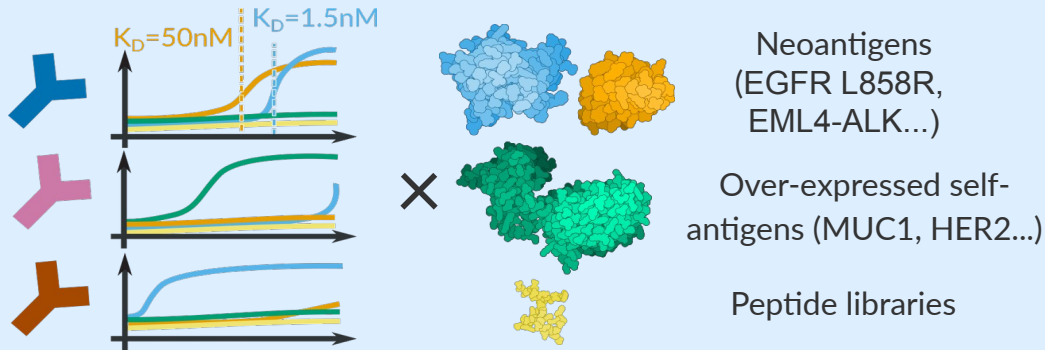
Over-expressed self-
antigens (MUC1, HER2...)

Peptide libraries

Which antigens are targeted by TIL-Bs



Measure the specificity of the repertoire *ex-vivo* (WP1/WP3)



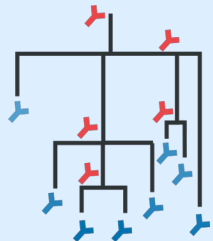
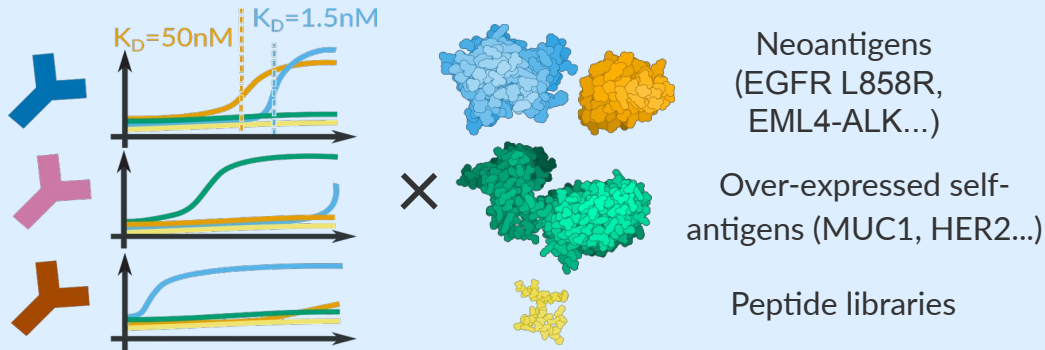
Ancestral
reconstruction

Did the antibody evolve
against this antigen ?

Which antigens are targeted by TIL-Bs



Measure the specificity of the repertoire *ex-vivo* (WP1/WP3)



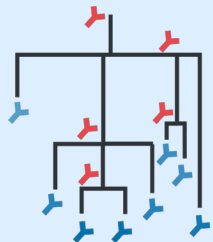
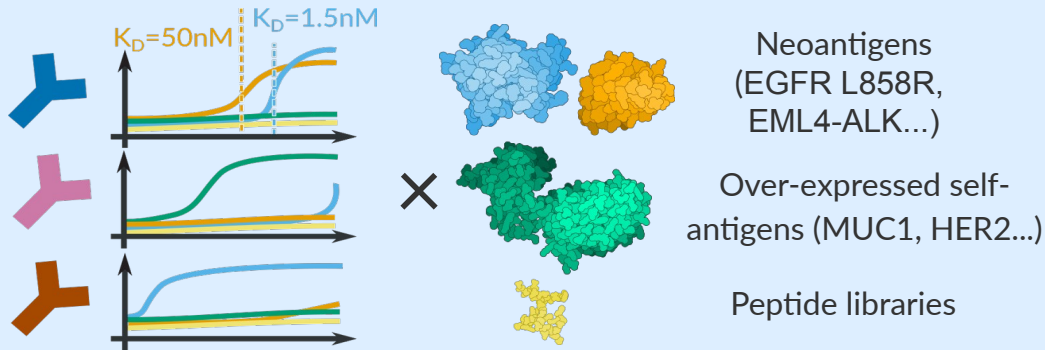
Ancestral reconstruction
Did the antibody evolve against this antigen?

Comparison with sera / PBMC
Pioneer / LUCA-PI Cohort
PhiP-Seq

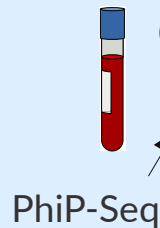
Which antigens are targeted by TIL-Bs



Measure the specificity of the repertoire *ex-vivo* (WP1/WP3)



Ancestral reconstruction
Did the antibody evolve against this antigen ?

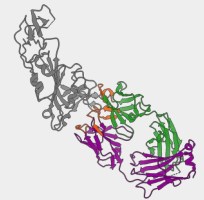


Comparison with sera / PBMC
Pioneer / LUCA-PI Cohort



Models & learning (WP2)

Generalize Ab-Ag predictions



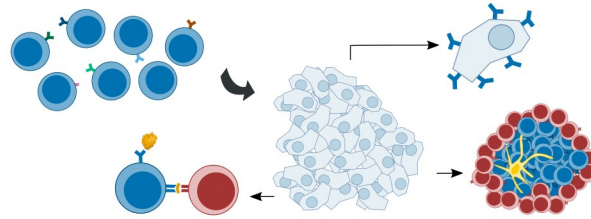
Integrate specificity to prognosis
A person icon followed by a large question mark.

THOMAS DUPIC

Affinity landscape of intra-tumoral B-cells in lung cancer

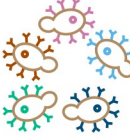


Project summary



Which antigens do intra-tumoral B cells target?

- BCRs as biomarkers
- New actionable antigens ?
- Plasma cell specificity
- Evolution of specificity

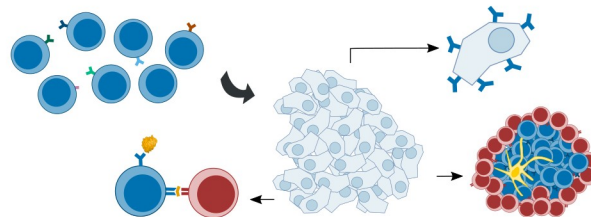


THOMAS DUPIC

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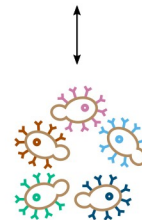


Project summary

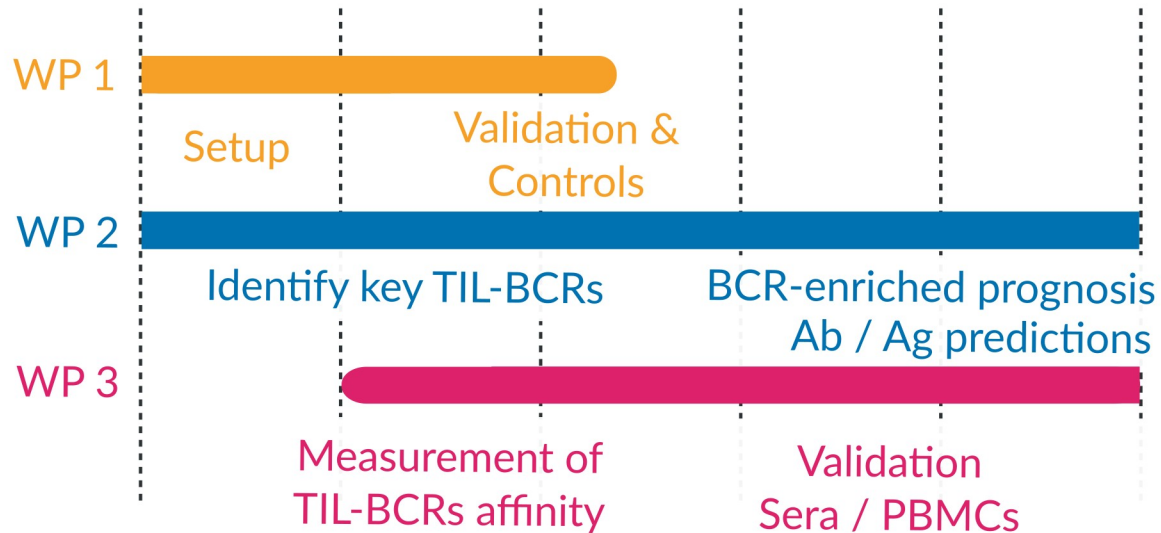


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Timeline



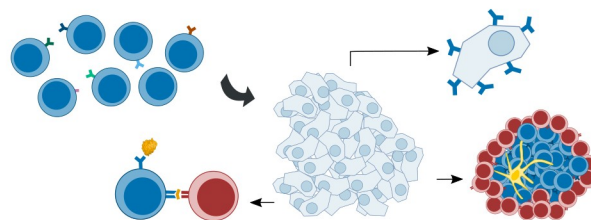
Post-doctoral researcher

THOMAS DUPIC

Affinity landscape of intra-tumoral B-cells in lung cancer

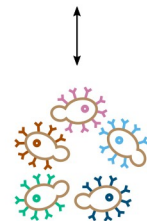


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Partner institutions

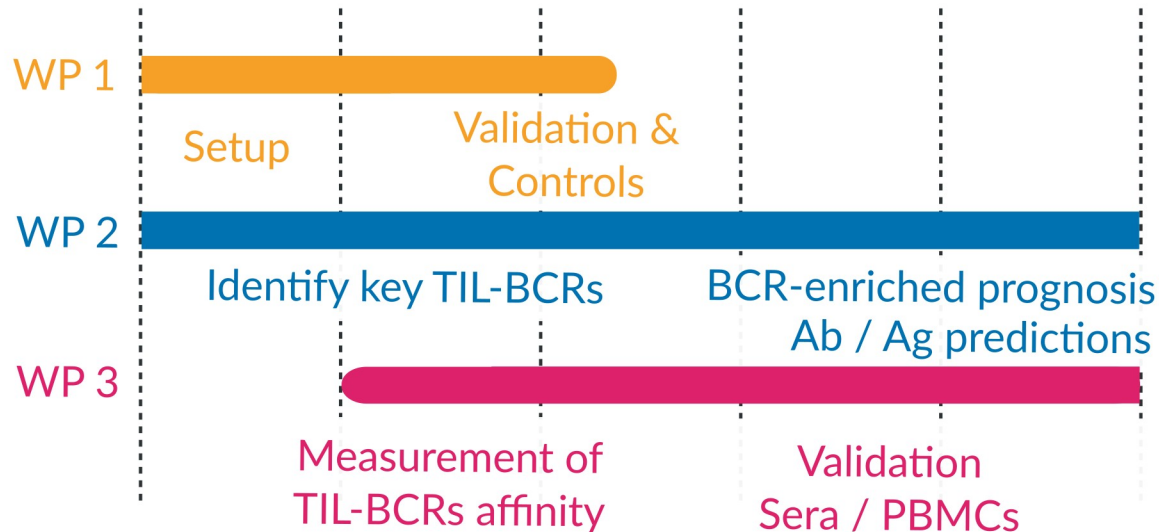


Co-financing
Idex $\Rightarrow$ 1 PhD



Team leader
Equipment,
lab, technician

Timeline



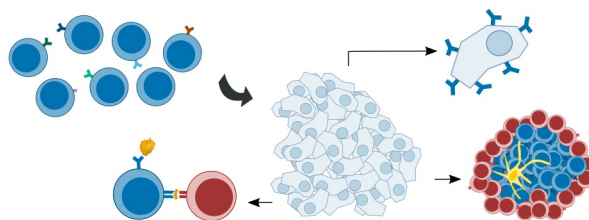
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THOMAS DUPIC

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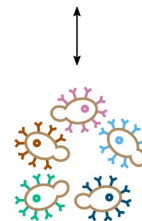


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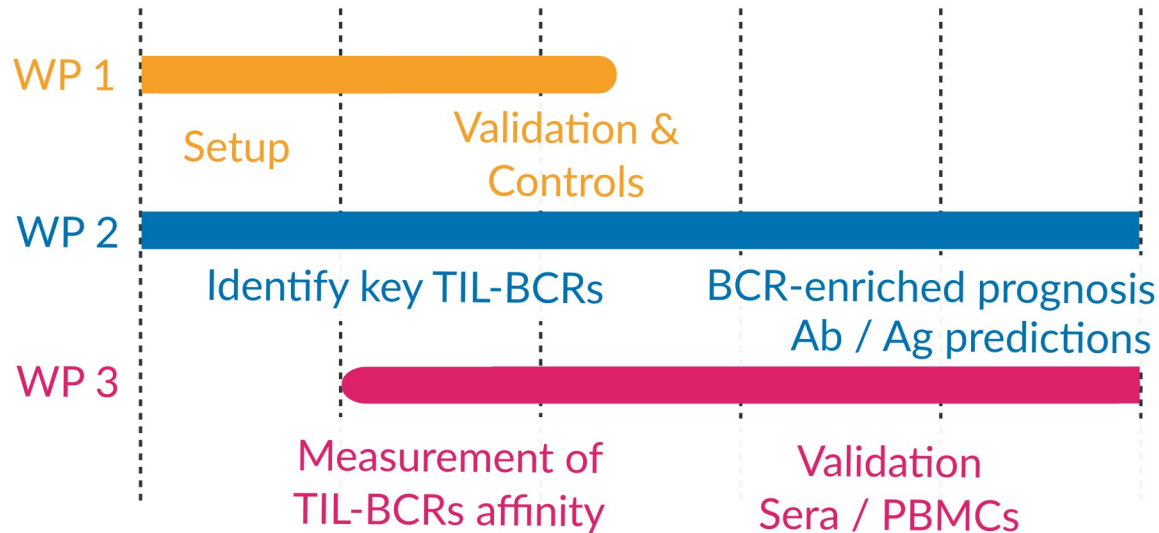


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Timeline



Post-doctoral researcher

Partner institutions

Inserm

amU
Aix Marseille Université

Co-financing
Idex $\Rightarrow$ 1 PhD

ap. hm

Cimi
IMMUNOLOGY

Team leader
Equipment,
lab, technician

Collaborations

CRCM
Centre de Recherche
en Cancérologie de Marseille

UCSF
 LPENS
LABORATOIRE DE PHYSIQUE
DE L'ÉCOLE NORMALE SUPÉRIEURE

P. Milpied
Q. Marcou
A. Phillips
T. Mora, A. Walczak

LUCA-PI

THE PIONEER PROJECT